```
C:\Users\DELL>arp -a

Interface: 192.168.56.1 --
  Internet Address      Ph
  192.168.56.255        ff
  224.0.0.2             01
  224.0.0.22            01
  224.0.0.251           01
  224.0.0.252           01
  224.0.0.253           01
  224.0.1.60            01
  239.0.0.8             01
  239.192.152.143       01
  239.255.67.250        01
  239.255.102.18        01
```

```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
C:\Users\DELL> netstat -a

Active Connections

  Proto  Local Address
  TCP    0.0.0.0:135
  TCP    0.0.0.0:445
  TCP    0.0.0.0:3306
  TCP    0.0.0.0:5040
  TCP    0.0.0.0:33060
  TCP    0.0.0.0:49664
  TCP    0.0.0.0:49665
  TCP    0.0.0.0:49666
  TCP    0.0.0.0:49667
  TCP    0.0.0.0:49668
  TCP    0.0.0.0:49676
  TCP    0.0.0.0:49677
  TCP    127.0.0.1:27017
  TCP    127.0.0.1:49671
  TCP    127.0.0.1:49672
  TCP    127.0.0.1:49673
  TCP    127.0.0.1:49674
```

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Exp no: 1	Ex no: 1
16/01/2026	BASIC NETWORKING COMMANDS

AIM:

To implement Basic Networking Commands in linux and Windows Operating system

BASIC NETWORKING COMMANDS:

- **arp -a:**

  ARP is short form of address resolution protocol.

- **hostname:**

  This is the simplest form of all TCP/IP commands.

- **ipconfig /all:**

  This displays detailed configuration information about connection including Router, Gateway, DNS, DHCP

- **nbtstat -a:**

  This command helps to solve problems with NETBIOS name resolution.

- **netstat**

  This is a tool used to perform DNS lookups in DNS details.

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```
ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [ME

  -f            Clears the routing tables of all
                used in conjunction with one of
                cleared prior to running the com

  -p            When used with the ADD command,
                boots of the system. By default,
                when the system is restarted. Ig
                which always affect the appropri

  -4            Force using IPv4.

  -6            Force using IPv6.

C:\Users\DELL> hostname
RISH-PC
C:\Users\DELL> ipconfig

ndows IP Configuration


nernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

nernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc4
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

reless LAN adapter Local Area Connection* 9:
```

```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP
        [-r] [-R] [-RR] [-s] [-S]

  -a   (adapter status) Lists the
  -A   (Adapter status) Lists the
                        IP addres
  -c   (cache)          Lists NBT
  -n   (names)          Lists loc
  -r   (resolved)       Lists nam
  -R   (Reload)         Purges an
  -S   (Sessions)       Lists ses
  -s   (sessions)       Lists ses
                        addresses

  -RR  (ReleaseRefresh) Sends Name

  RemoteName    Remote host machin
  IP address    Dotted decimal repr
  interval      Redisplays selecte
                between each displa
```

```
ers\DELL>pathping

: pathping [-g host-lis
           [-p period]
           [-4] [-6] ta


ns:
g host-list      Loose s
h maximum_hops   Maximum
i address        Use the
n                Do not
p period         Wait pe
q num_queries    Number
w timeout        Wait ti
4                Force u
6                Force u
```

- nslookup :
  - Operate
- Pathping
  - basically
- Ping :
- Route
  - the

LINUX

- ip
  - ma
  - de
- ifc

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- **nslookup** :

Used to perform DNS lookups . nslookup can operate in two modes interactive, non - interactive

Eg. www.google.com

- **Pathping** :

Pathping is a unique to Window's and is basically a combination of Tracert commands

- **Ping** :

Ping is a command to test connectivity nodes

- **Route** :

Route command is used to show / manipulate the IP routing table.

LINUX NETWORKING COMMANDS :

- **ip** :

The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels

- **ifconfig** :

The ifconfig command was a staple in many sysadmins tool belt for config and troubleshooting

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